Esha Gupta

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Education

Carnegie Mellon University, Heinz College, Pittsburgh, PA **2026 – 2028 (expected)**
M.S. in Public Policy and Management, Data Analytics QPA: 4.0/4.0

Indian Institute for Human Settlements, Bangalore, India **2022 – 2023**
Urban Fellows Program

Jai Hind College, Mumbai, India **2018 – 2021**
B.A. Economics, Minor in Political Science CGPA: 9.67/10.0

Skills & Coursework

Technical Skills Python, Excel, R, SQL, QGIS, Datawrapper, LLM benchmarking, LaTeX, Prompt Engineering

Coursework Intelligent Transportation Systems, Statistical Reasoning in R, Exploratory Data Analysis in Python, Machine Learning Foundations with Python, Optimization and Decision Modeling for Analytics, Econometrics (Spring 2027)

Projects

- **MBTA Fare-Free Transit Analysis | Boston, MA — CMU, 2026** Combined MBTA ridership, passenger survey, and ACS demographic data to compare fare-free and regular-fare bus routes. Conducted exploratory analysis of ridership trends, passenger characteristics, and neighborhood accessibility, visualizing differences across routes to evaluate transit policy outcomes.
- **Mapping KR Puram:** Conducted fieldwork to map and analyze the impact of an upcoming metro station on the **economic, ecological, and infrastructural vulnerability** in neighboring communities. Research informed stakeholder meetings with relevant government agencies.
- **CMU Build-a-Thon 2026: MindTrace:** Built a website hosting a collection of mental exercises for people aged 50+ to track fluctuations in their memory and cognitive abilities. **Top 6 finalists of 300 participants.**

Work Experience

Animal Law & Policy Network **Feb 2026 – Mar 2026**
Research Associate

- Surveyed **400+** non-governmental organizations (NGO) and interviewed **18 of 35 NGOs, 3 activists and 2 government officials** to catalog active street dog welfare programs in Mumbai, India.
- Applied **qualitative coding** across all 35 interviews to identify proxy indicators, then designed and administered a structured survey back to them to recommend funding opportunities to granting agencies.
- Conducted **service accessibility analysis** of NGO welfare resources on **QGIS** through heatmaps, isochrone maps, and point-maps to identify potential funding opportunities.

Independent Research **May 2025 – Nov 2025**
Applied ML Researcher

- Mentored by **Dr. Kingsuk Maitra, Qualcomm** to map water pollution in Indian rivers using machine learning, geospatial analysis, and water quality analysis.
- Tested fine-tuning workflows with **SmolVLM** to predict water turbidity from satellite imagery, iterating across training configurations, dataset sizes, epoch counts, and input formats.
- Built a dataset of **3,000+ multispectral points**, 29 satellite image patches, and 1,258 raw GPS-tagged water-quality readings using Excel and QGIS.
- Designed fine-tuning instructions and benchmarked inference strategies, including hyperparameter tuning and model evaluation.
- Evaluated model performance using MAE and MSE across experiments, **reducing MAE from 0.52 to 0.17** through iterative adjustments to training sample size and epoch count.

University of Chicago Trust **Jan 2024 – Apr 2025**
Research Associate

- Supported **3 water quality projects** across LLM research, field-based pollution mapping, and AR-based public engagement.
- WQ-Info: Conducted **literature reviews** on fine-tuning methods, bias and hallucination mitigation, and accuracy reward models; benchmarked commercial LLMs for extracting unstructured PDF data.
- Dehradun Water Mapping: Used sensor technology to map pollution across a 5 km river site; analyzed fieldwork data in Excel, Python, and QGIS to examine spatial pollution patterns through **point maps, binned maps, heatmaps, violin plots, and scatterplots.**
- Stakeholder Research: Documented **stakeholder conversations**, identified collaboration opportunities, and helped define user needs for water-quality tools for public health impacts.
- WQ-Sandbox v2: Developed a project plan and coordinated consultants/vendors for an AR-based exhibition for 200 students aged 10-14, curating a user journey on river pollution dispersion.

Artha Global **Apr 2023 – Sep 2023**
Research Intern

- Supported **4 projects on urban analysis:** Clean Air Zones, climate-resilient transportation infrastructure, urban-planning capacity building for government officials, nature-based solutions, and wastewater management in urban India.
- Performed secondary research, collected data, documented case studies, reviewed literature, **authored 3 project reports, 2 concept notes, and 1 proposal**, created maps, and drafted capacity-building modules.

Content People AB **Oct 2022 – Jun 2026**
Project Manager

- Coordinated senior stakeholders and partners across India, Sweden, and Germany to organize **6 events** on sustainability, green transition, and women in STEM with **150+ cumulative attendees.**
- Oversaw end-to-end website development and **led 10+ person teams** producing **50+ public-facing and speaker-facing materials**, including social media assets, guest briefs, event documents, and logistics materials.

ICRIER **Oct 2021 – Dec 2021**
Research Intern

- Conducted field research for **Indian Council for Research on International Economic Relations** to evaluate the "Pradhan Mantri Awas Yojana" rental housing scheme in Mumbai.
- **Visited 12 sites** built under the scheme and conducted **25+ consultations** with beneficiaries and government officials to evaluate access to transport, utilities, and economic activities.
- Conducted policy research on business models of affordable housing and co-living institutions in India, informing a paper on housing for low-income urban households.

Publications & Writing

- Dhamelia, Shubham, **Gupta, Esha**, Rajani, Sivani, & Joshi, Shubham. Mapping Community Dog Welfare NGOs in Mumbai Suburban and Thane: Assessing Capacities, Gaps, and Opportunities. *ALPN Research Foundation*, Bengaluru, 2026.
- Roy, D., Ml, M., Haque, I., & **Gupta, E.** Institutional Provisioning of Affordable Rental Housing: Past Experience & Way Forward. *ICRIER*, New Delhi, 2022.
- The Cost of Incognizance: How Loneliness Affects Productivity. *Harvard Economics Review*, Jul 2020. First-place winner on the Economics of Loneliness.